Algorithm Analysis And Data Structures

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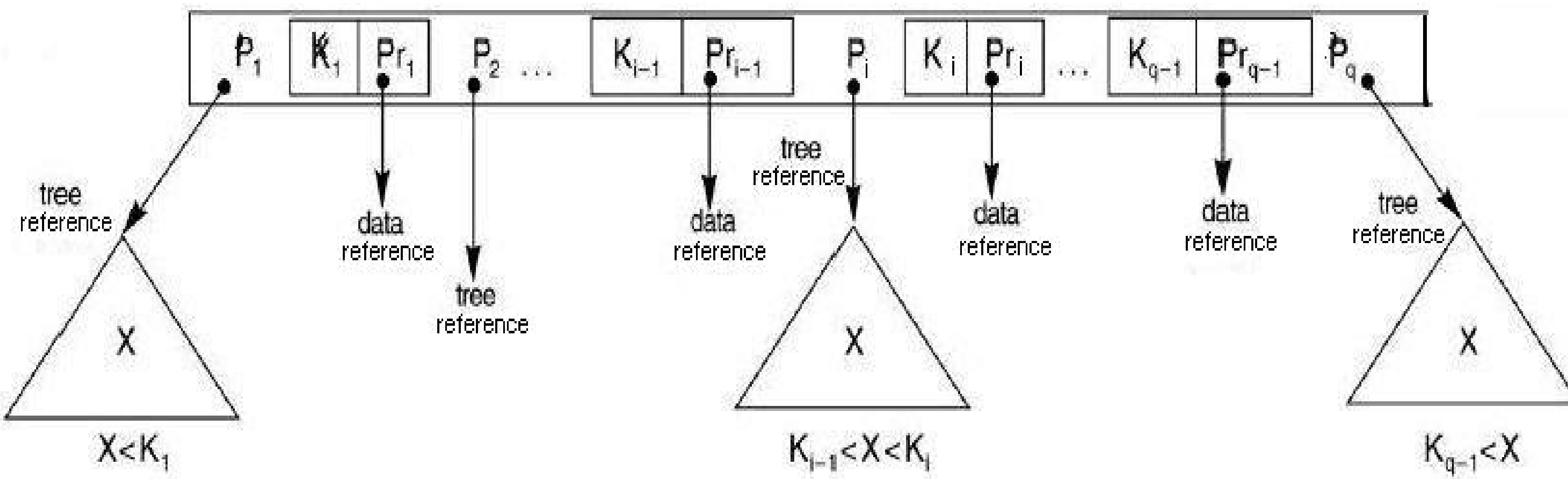
Description of B+ Trees

- A variation of B-Trees
- B-Trees commonly found in databases and filesystems
- Sacrifices space for efficiency (not as much re-balancing required compared to other balanced trees)

Description of B+ Trees (cont.)

- Main difference of B-Trees and B+ Trees
  - B-Trees: data stored at every level (in every node and leaves)
  - B+ Trees: data stored only in leaves
  - Internal nodes only contain keys and index/tree pointers in B+
  - All leaves are at the same level (the lowest one)

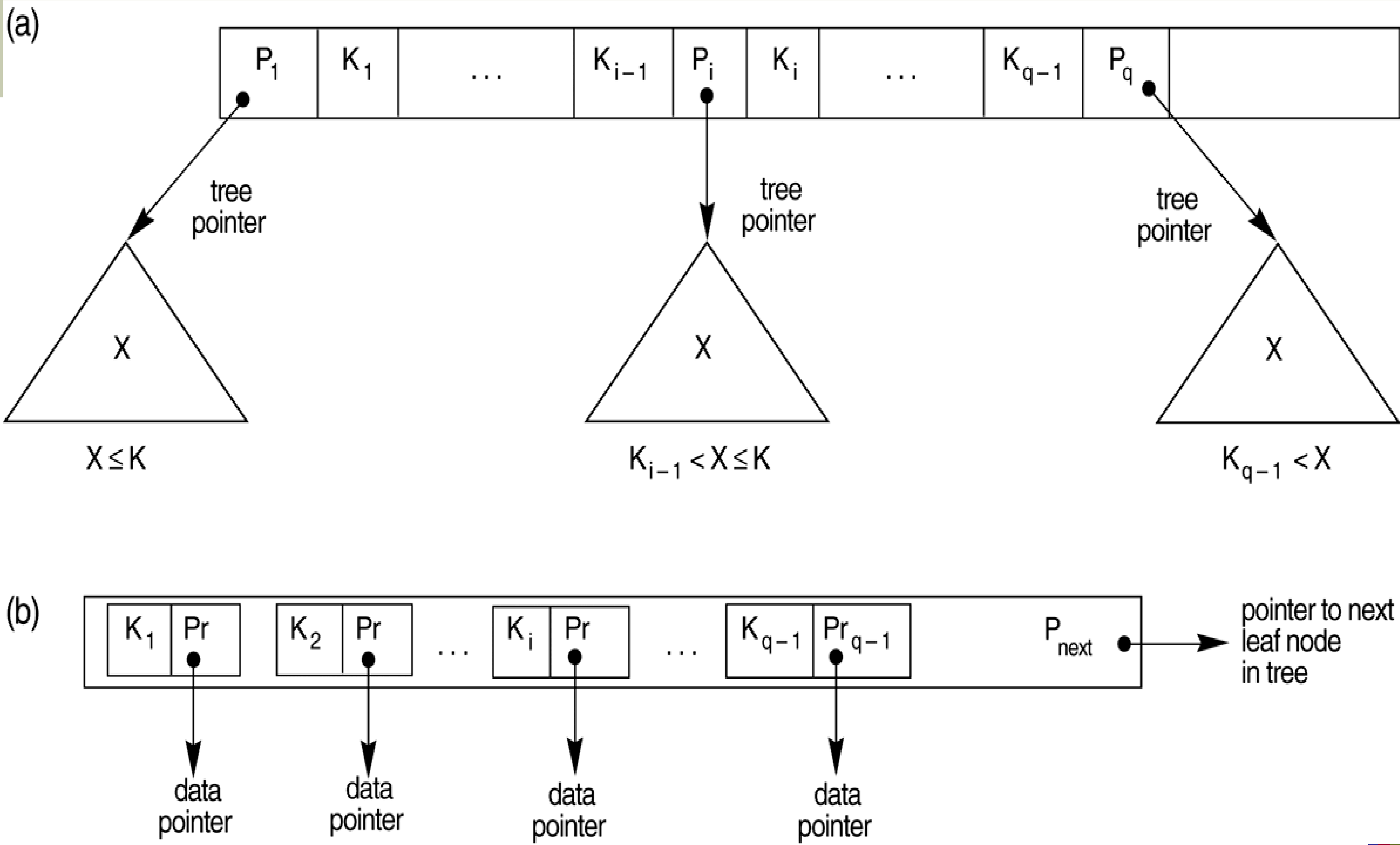
The node structure of a Multi-way tree



Note:

- Corresponding to each key there is a data reference that refers to the data record for that key in secondary memory.
- In our representations we will omit the data references.
- The literature contains other node representations that we will not discuss.

The node structure of a B+ tree



Description of B+ Trees (cont.)

- B+ trees have an order n
- An internal node can have up to n-1 keys and n pointers
- Built from the bottom up

Description of B+ Trees (cont.)

- All nodes must have between ceil(n/2) and n keys (except for the root)

- For a B+ tree of order n and height h, it can hold up to n^h keys

Searching in B+ Trees

- Searching just like in a binary search tree
- Starts at the root, works down to the leaf level
- Does a comparison of the search value and the current “separation value”, goes left or right

Inserting into a B+ Tree

- A search is first performed, using the value to be added
- After the search is completed, the location for the new value is known

Inserting into a B+ Tree (cont.)

- If the tree is empty, add to the root
- Once the root is full, split the data into 2 leaves, using the root to hold keys and pointers
- If adding an element will overload a leaf, take the median and split it

Inserting into a B+ Tree (cont.)

- **Example: Insert the keys of 8, 5, 1, 7, 3, 12, 9, 6 into 3 order B+ tree.**

Suppose we had a B+ tree with n = 3

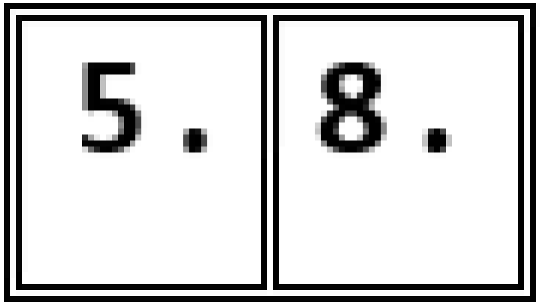
- 2 keys max. at each internal node
- 3 pointers max. at each internal node
- Internal nodes include the root

B+- Tree order 3 Insertion

- Inserting value 8
- Since the node is empty, the value must be placed in the leaf node.

B+- Tree Insertion Cont.

- Inserting value 5
- Since the node has room, we insert the new value.



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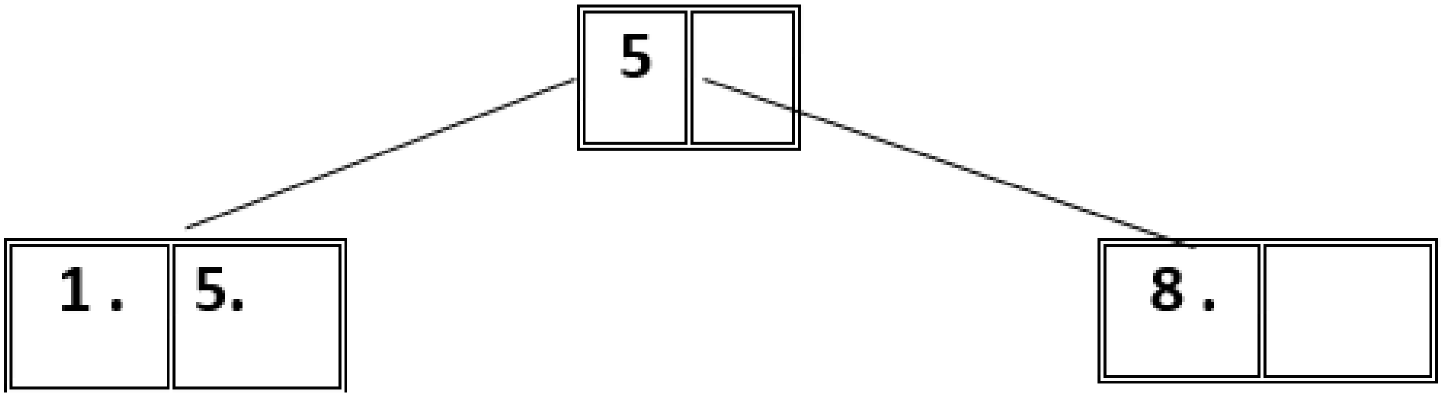
B+- Tree Insertion Cont.

- Insert value 1
- Since the node is full, it must be split into two nodes.
- Each node is half full.



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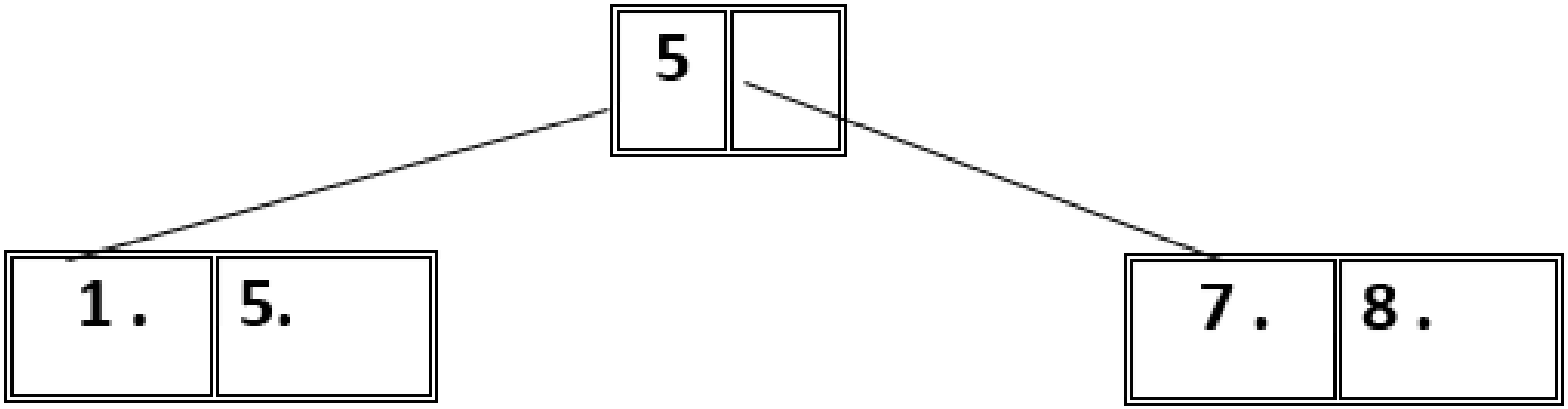
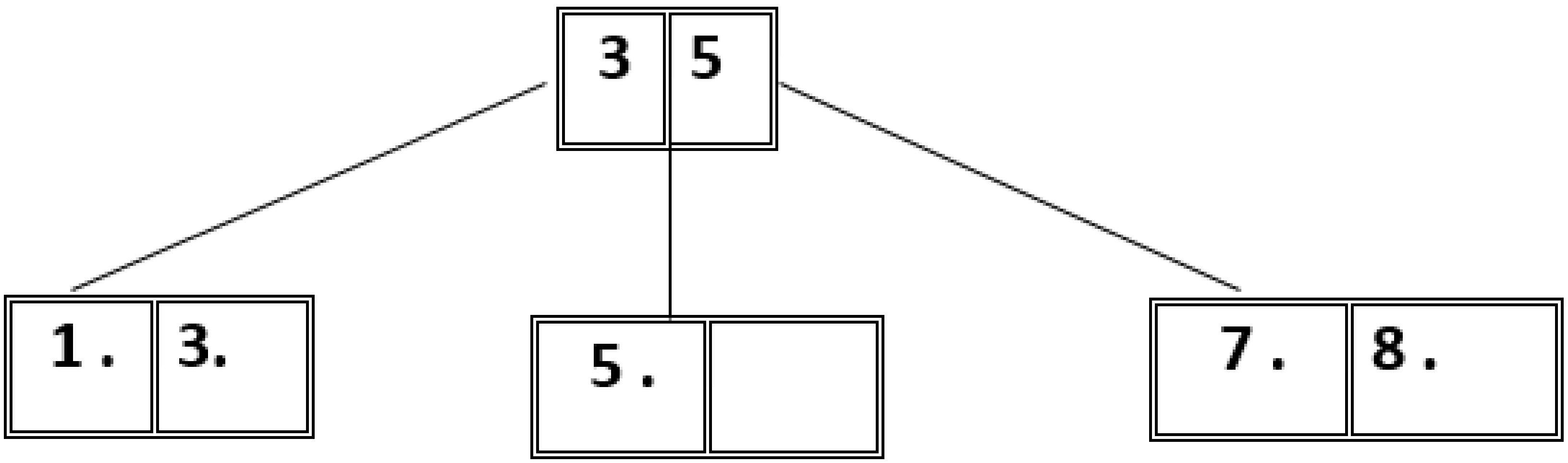
B+- Tree Insertion Cont.

- Inserting value 7.



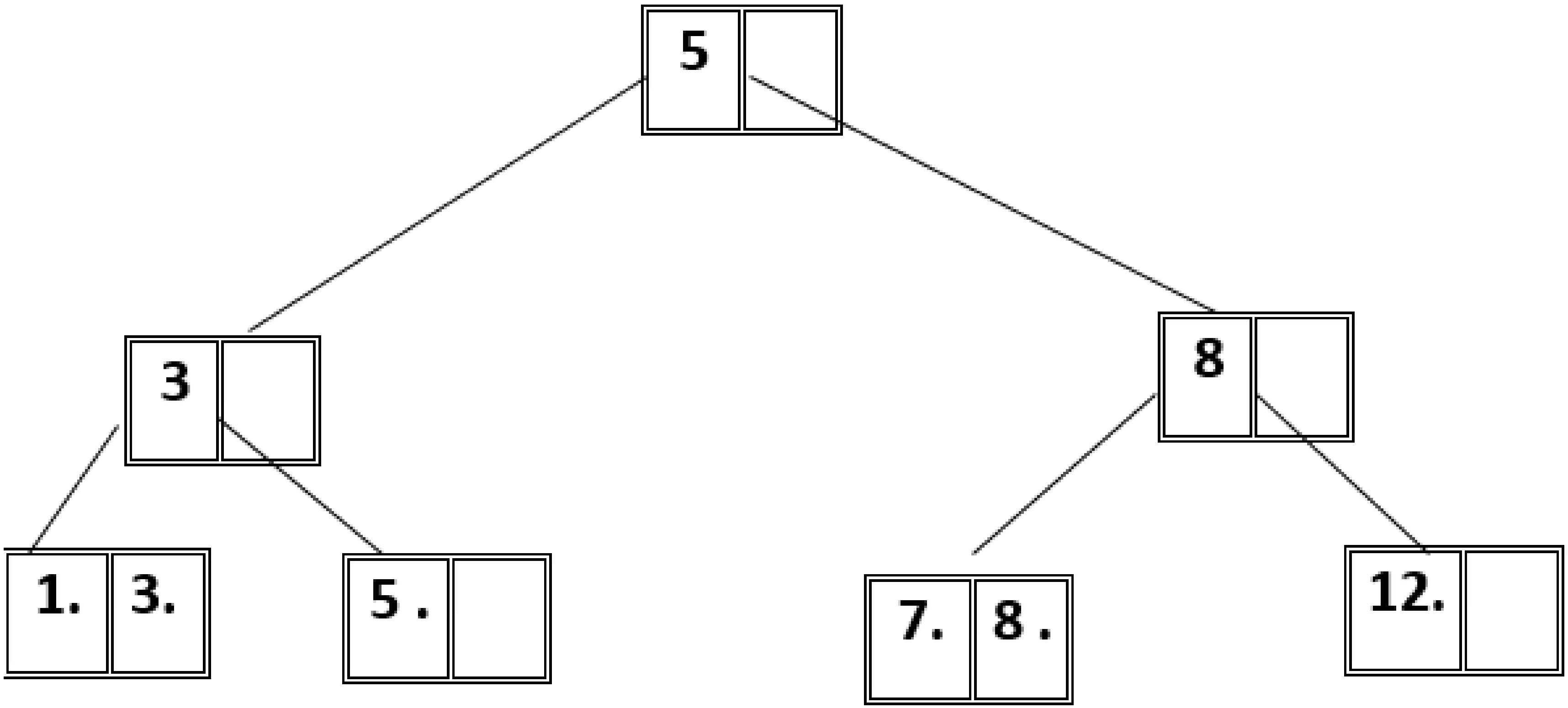
B+- Tree Insertion Cont.

- Inserting value 3



B+- Tree Insertion Cont.

- Inserting value 12



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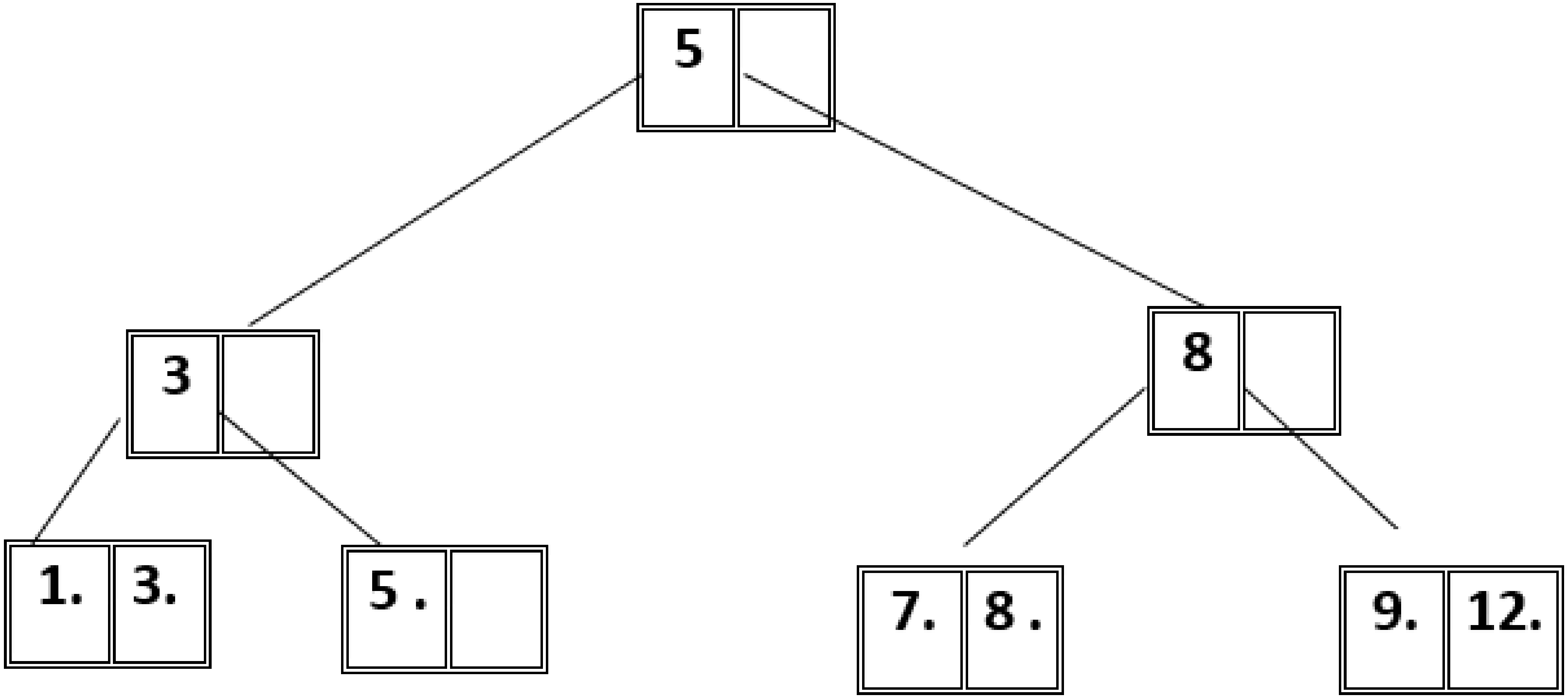
B+- Tree Insertion Cont.

- Inserting value 9



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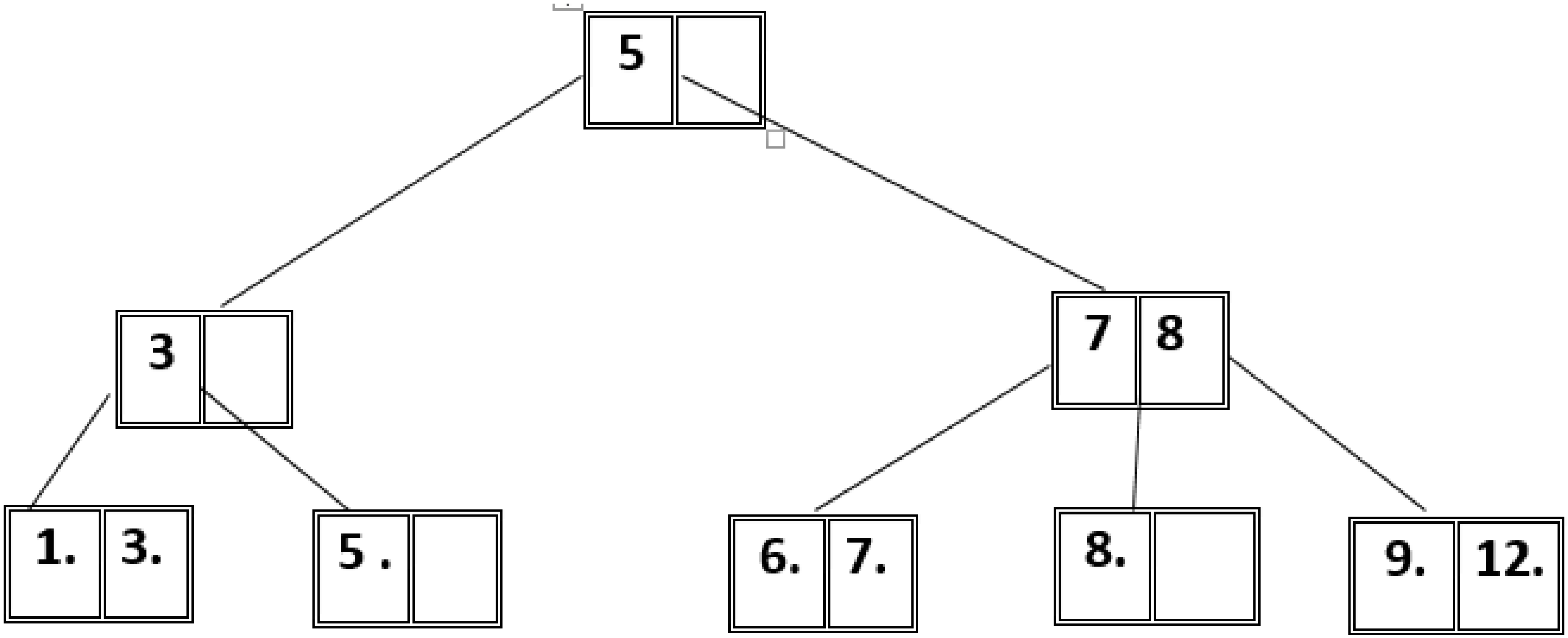
B+- Tree Insertion Cont.

- Inserting value 6



Example

- Insert the following keys to a 3-order B+ tree

89,78,8,19,20,33,56,44

(Solved in the class)

- Insert the following keys to a 3-order B+ tree

  44,56,33,20,19,8,78,89

(Solved in the class)

Example

- Insert the following keys to a 4-order B+ tree

  92,24,6,7,11,8,22,4,5,16,19,20,78

(Solved in the class)

Exercise

- Insert the following keys to a 5-order B+ tree

  92,24,6,7,11,8,22,4,5,16,19,20,78

- Insert the following keys of 100,110,120,130, 140, 150,90,80,70,60,50,51,52,53,129,128,127,126,125,
124,131,132,133,134,135,136,137,138,170,169,168,
167,166,165,164,163,162,161,160,159,158,157

  a) To 5 order B+ tree b) To 6 order B+ tree

  c) To 7 order B+ tree d) To 5 order B tree

  e) To 6 order B tree f) To 7 order B tree

Questions and Answers

Thank you ...

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